# Notes: Bond, Goldstein, and Prescott (RFS, 2010)
## Market-Based Corrective Actions

## Contents

# 1 One-sentence summary

Bond, Goldstein, and Prescott study situations in which an agent uses a market price to decide whether to take a corrective action that itself changes the security's value; the key problem is that the price both informs and anticipates the action, so market-based intervention can become self-defeating unless the agent has sufficiently precise private information or observes additional informative market signals.

# 2 Big picture

- **Question.** When can boards, activists, regulators, or managers use financial-market prices to guide corrective actions?
- **Core problem.** Prices summarize information about fundamentals, but they also reflect expected future corrective actions. Therefore the price is not a passive signal: it is partly an equilibrium object created by the very action it is meant to guide.
- **Main feedback loop.** A low fundamental may trigger intervention. Intervention improves value. The price may then rise. A moderate price can therefore mean either "fundamental is moderate and no intervention" or "fundamental is bad but intervention is expected."
- **Main theoretical difficulty.** When intervention is value-improving at the cutoff, the price function is nonmonotone in fundamentals. A single price can correspond to fundamentals on both sides of the intervention threshold.
- **Main result.** Market prices and the agent's own information are complements. The agent needs private information to interpret whether the market price reflects fundamentals or anticipated intervention.
- **Security design matters.** Convex securities, such as equity, can support equilibria with too much intervention. Concave securities, such as debt, can support equilibria with too little intervention.
- **Policy message.** Market discipline cannot simply replace direct supervision or private monitoring. Market prices are useful only when the decision maker can interpret them with enough independent information.

# 3 Incremental contribution of the paper

## 3.1 Endogenizing the informational content of prices

The paper studies a feedback setting in which prices are informative because they reflect market information, but the same prices are distorted by expectations about the agent's future action. This differs from standard rational-expectations pricing models in which the value of the security is exogenous to the action taken after observing the price.

## 3.2 A nonmonotone price function as the central object

A central contribution is the analysis of the empirically relevant case in which corrective intervention raises firm value. In that case, the price can jump upward exactly when fundamentals become bad enough to justify intervention. This creates a nonmonotone price function and makes inference from prices fundamentally hard.

### 3.3 Equilibrium analysis of self-defeating market signals

The paper goes beyond noting that market-based policies may destroy the information in prices. It characterizes when rational expectations equilibria exist, when the agent can implement his preferred intervention rule, and when equilibrium multiplicity or nonexistence arises.

### 3.4 Complementarity between market information and agent information

A key insight is that better agent information makes market prices more useful. This is not because the market is uninformed; in the model, the market observes the fundamental. Rather, the agent's own signal helps him separate fundamental information from expectations about his own intervention.

### 3.5 Positive and normative implications

The paper provides guidance for empirical research on governance and regulation. If prices both affect and reflect intervention, regressions of actions on prices face dual causality. The paper also suggests policy tools: observe multiple securities, disclose the agent's information, or create a prediction market tied directly to intervention.

## 4 Model setup

### 4.1 Timing

The model has one firm, one agent, and a financial market that trades a security of the firm. There are three dates:

- **Date 0.** The market price of a security is determined.
- **Date 1.** The agent observes the price and a private signal, then decides whether to intervene.
- **Date 2.** Security holders are paid based on realized firm cash flows.

### 4.2 Firm cash flow and fundamentals

Absent intervention, firm assets produce gross cash flow

$$y = \theta + \delta, \tag{1}$$

where $\theta$ is the fundamental, drawn at date 0, and $\delta$ is residual uncertainty realized at date 2. The residual shock has mean zero.

The fundamental $\theta$ is the key state variable. High $\theta$ means the firm is healthy; low $\theta$ means the firm is weak and may benefit from corrective action.

### 4.3 Security value without intervention

Let $X(\theta)$ denote the value of the traded security absent intervention. For equity with debt face value $D$, the value is

$$X(\theta) = \int_{D-\theta}^{\infty} (\theta + \delta - D) f(\delta) d\delta. \tag{2}$$

For equity, $X'(\theta) > 0$ and $X''(\theta) > 0$, so the security value is increasing and convex in fundamentals.

**Interpretation.** Equity behaves like a call option on the firm's cash flow. Its value rises with fundamentals and becomes more sensitive at high fundamentals because downside losses are limited by debt.

### 4.4 Intervention technology

At date 1, the agent can intervene. Intervention changes expected cash flow by $T(\theta)$. The paper assumes:

- $T(\theta)$ weakly decreases in $\theta$;
- $\theta + T(\theta)$ increases in $\theta$.

The first assumption says intervention is more valuable when fundamentals are worse. The second says that even after intervention, better fundamentals still imply better total expected cash flow.

The effect of intervention on the traded security is

$$U(\theta) = X(\theta + T(\theta)) - X(\theta). \tag{3}$$

This is the incremental security value from intervention.

### 4.5 Agent's payoff from intervention

The agent bears a private cost $C > 0$ if he intervenes. His gross benefit is $V(\theta)$. The paper assumes that there is a unique cutoff $\hat{\theta}$ such that

$$V(\theta) \geq C \quad \text{if and only if} \quad \theta \leq \hat{\theta}. \tag{4}$$

Thus a fully informed agent would intervene exactly when fundamentals are sufficiently low.

### 4.6 Information and intervention policy

The market observes $\theta$ and sets the price rationally. The agent does not directly observe $\theta$. Instead, he observes a noisy signal

$$\phi = \theta + \xi, \tag{5}$$

where $\xi$ is uniformly distributed on $[-\kappa, \kappa]$. The parameter $\kappa$ measures the imprecision of the agent's private information: lower $\kappa$ means a more precise signal.

The agent chooses an intervention policy

$$I(P, \phi) \in [0, 1], \tag{6}$$

where $P$ is the observed market price.

### 4.7 Rational expectations pricing

In equilibrium, the price reflects the security value given fundamentals and the expected intervention probability:

$$P(\theta) = X(\theta) + E_\phi \left[ I(P(\theta), \phi) \mid \theta \right] U(\theta). \tag{7}$$

The price incorporates both information about fundamentals and expectations about the agent's response to the price.

### 4.8 Best response condition

Given a price $\widetilde{P}$ and a signal $\widetilde{\phi}$, the agent intervenes when the expected benefit exceeds the cost:

$$I(\widetilde{P}, \widetilde{\phi}) = \begin{cases} 1, & \text{if } E[V(\theta) \mid P(\theta) = \widetilde{P}, \widetilde{\phi}] > C, \\ 0, & \text{if } E[V(\theta) \mid P(\theta) = \widetilde{P}, \widetilde{\phi}] < C. \end{cases} \tag{8}$$

### 4.9 Equilibrium definition

An equilibrium is a pricing function $P(\cdot)$ and an intervention policy $I(\cdot, \cdot)$ satisfying both rational expectations pricing and the agent's best response. The equilibrium concept is rational expectations: the price must correctly forecast the expected action, and the action must be optimal given the information contained in the price and the signal.

## 5 Equilibrium logic

### 5.1 Agent-preferred equilibrium

An agent-preferred equilibrium is one in which the agent implements the same intervention rule he would use if he knew the fundamental:

$$I = 1 \quad \text{if } \theta < \hat{\theta}, \qquad I = 0 \quad \text{if } \theta > \hat{\theta}. \tag{9}$$

Under this rule, the price function is

$$P(\theta) = \begin{cases} X(\theta + T(\theta)), & \theta < \hat{\theta}, \\ X(\theta), & \theta > \hat{\theta}. \end{cases} \tag{10}$$

This expression is the central object in the paper. It shows why the equilibrium price can fail to reveal the fundamental.

### 5.2 Monotone price function: $T(\hat{\theta}) \leq 0$

If intervention at the cutoff reduces expected security value, the price function is monotone. Each price corresponds to exactly one fundamental. The agent can therefore infer $\theta$ from the price alone, regardless of the noise in his own signal.

**Proposition 1.** If $T(\hat{\theta}) \leq 0$, then for every signal accuracy $\kappa$, an equilibrium with agent-preferred intervention exists and is unique.

**Economic interpretation.** This is the simple case. The price is fully revealing because intervention does not create an upward price jump at the cutoff. This case resembles standard feedback models in which market prices remain monotone and easy to interpret.

### 5.3 Nonmonotone price function: $T(\hat{\theta}) > 0$

The main analysis focuses on corrective actions, where intervention raises expected cash flow at the cutoff. Then $T(\hat{\theta}) > 0$, and the agent-preferred price function is nonmonotone. As fundamentals fall below the cutoff, the agent intervenes, and the expected payoff of the security rises. Therefore a bad fundamental with expected intervention can have the same price as a better fundamental without intervention.

**Inference problem.** A single price can be consistent with two states:

- a low $\theta$ state in which intervention is expected; or
- a higher $\theta$ state in which intervention is not expected.

The agent's private signal is needed to distinguish these states.

### 5.4 Role of signal precision

The paper's key comparative static is in $\kappa$, the noise in the agent's private signal.

**Proposition 2.** If $\kappa < T(\hat{\theta})/2$, then an agent-preferred equilibrium exists. It is unique when $\kappa$ is sufficiently small. When $\kappa$ is close to $T(\hat{\theta})/2$, additional equilibria can exist. If $\kappa > T(\hat{\theta})/2$, no agent-preferred equilibrium exists, and for sufficiently large $\kappa$ no rational expectations equilibrium exists.

**Interpretation.**

- When the agent's signal is precise, he can distinguish the two fundamentals that share a price.
- When the signal is moderately precise, an agent-preferred equilibrium may coexist with other equilibria.
- When the signal is imprecise, the agent cannot reliably infer whether a price reflects fundamentals or anticipated intervention.

### 5.5 Equilibria with too much intervention

For convex securities such as equity, equilibria without agent-preferred intervention can feature too much intervention.

**Proposition 3.** When $\kappa < T(\hat{\theta})/2$ is sufficiently close to $T(\hat{\theta})/2$, equilibria exist in which the agent intervenes with positive probability for some fundamentals above $\hat{\theta}$. Moreover, for a convex security, any equilibrium other than the agent-preferred equilibrium entails positive intervention at some $\theta > \hat{\theta}$.

**Economic interpretation.** If the agent sometimes intervenes even when fundamentals are above the cutoff, this raises the price in those states. The altered price makes those states look more like low-fundamental states where intervention should occur. The agent then has weaker evidence against intervention, making the overintervention self-confirming.

### 5.6 Concave securities and too little intervention

The paper also discusses how the result changes for concave securities, such as debt in some relevant ranges. For a concave security, equilibria without agent-preferred intervention can instead feature too little intervention.

**Economic interpretation.** Equity payoff convexity amplifies upside effects of intervention, making overintervention equilibria possible. Debt payoff concavity dampens the upside and can support underintervention equilibria. Thus the type of security observed by the agent matters for the direction of the intervention bias.

# 6 Applications

## 6.1 Corporate governance

The model maps naturally to boards of directors and shareholder activists.

- $\theta$ is the firm's expected cash flow absent intervention.
- The agent is a board, director, activist, or other governance player.
- Intervention includes CEO replacement, pressure on management, or operational changes.
- The private cost $C$ includes reputational cost, effort, legal cost, or loss of private benefits.

**Interpretation.** Low market valuations may trigger governance action, but the price also reflects the expectation of such action. A moderate price after a bad signal need not mean the firm is fundamentally moderate; it may mean the market expects the board or activists to act.

## 6.2 Bank supervision

Bank supervisors can use market prices of equity or subordinated debt to assess bank health. In the model:

- $\theta$ captures bank fundamentals;
- intervention includes restrictions, liquidity support, asset guarantees, management changes, or closure;
- the supervisor's payoff may reflect total surplus or protection of depositors.

**Interpretation.** Bank security prices may become nonmonotone because bad fundamentals can trigger regulatory support or intervention. Hence the absence of a very low security price does not necessarily imply that the bank is healthy.

## 6.3 Managerial investment and acquisition decisions

Managers may learn from their own stock prices after announcing an acquisition or investment plan. In this setting:

- $\theta$ is the expected cash flow if the action proceeds;
- intervention means cancelling the action;
- $C$ captures private benefits or reputational costs of cancelling.

**Interpretation.** A manager who observes a negative stock-market reaction may infer that investors have adverse information about the project. But the market reaction itself may depend on beliefs about whether the manager will cancel the project.

# 7 Improving market-based learning

## 7.1 Multiple securities

If the agent observes both a convex and a concave security, such as equity and debt, the multiple-equilibrium problem can be reduced.

**Proposition 4.** If $\kappa < T(\hat{\theta})/2$ and the agent observes prices of both a strictly concave and a strictly convex security, then the agent-preferred equilibrium is unique. However, if the agent's information is too imprecise, no equilibrium still exists.

**Economic message.** Multiple securities provide different transformations of the same fundamentals. This helps the agent separate fundamentals from intervention expectations, but it does not solve the problem when the agent's own signal is very poor.

## 7.2 Transparency and disclosure

If the agent publicly discloses his signal before the market price forms, the price can become more revealing.

**Interpretation.** Disclosure eliminates some multiplicity because the market knows what the agent knows and can better forecast his action. But transparency may not solve the no-equilibrium problem when the agent's signal is too imprecise. In bank regulation, transparency can also reduce banks' willingness to share information with the regulator.

## 7.3 Prediction markets

The paper considers a security that pays one if the agent intervenes and zero otherwise.

**Proposition 5.** If both a standard equity security and a prediction-market security are traded, then for all $\kappa$ the unique equilibrium features agent-preferred intervention.

**Economic message.** A prediction market directly prices the probability of intervention. Combining this probability with the equity price allows the agent to infer fundamentals and recover the preferred intervention rule.

## 7.4 Commitment

Could the agent commit ex ante to an intervention rule based only on the price? The paper shows that this does not fully solve the problem.

**Proposition 6.** If the agent commits in advance to an intervention policy based on the price of one security, he cannot achieve his preferred intervention. The set of fundamentals at which he deviates from the preferred rule is at least of size $T(\hat{\theta})$.

**Interpretation.** Commitment makes equilibrium existence easier but cannot replicate the ideal rule. The reason is that the preferred rule itself creates a nonrevealing price function, so a price-only commitment cannot fully separate states.

# 8 Analytical design and assumptions

## 8.1 What drives the results

The core results rely on three elements:

- the agent wants to intervene at low fundamentals;
- intervention raises value near the cutoff;

- the price is forward looking and reflects expected intervention.

Together these assumptions create the nonmonotone price function.

### 8.2 Why the market can be perfectly informed and still hard to use

The market observes $\theta$ in the baseline model. Yet the agent cannot necessarily extract $\theta$ from the price, because the price is an equilibrium object that includes the market's expectation of the agent's own action. Thus the problem is not lack of market information; it is the feedback between market prices and intervention.

### 8.3 Security curvature

The distinction between convex and concave securities is central. Equity is convex because of limited liability. Debt can be concave in ranges where it is close to being paid in full. This curvature shapes whether the non-agent-preferred equilibrium involves too much or too little intervention.

### 8.4 Bounded signal noise

The uniform signal noise assumption gives the agent a bounded confidence interval for the fundamental. This tractability helps establish when the agent can separate two fundamentals with the same price. Economically, the key point is that the agent's own information must sometimes rule out one of the price-consistent states.

## 9 Empirical implications

### 9.1 Dual causality

If agents use market prices to decide whether to intervene, there is two-way causality:

- market prices affect intervention decisions;
- expected intervention decisions affect market prices.

Empirical work that regresses interventions on prices without accounting for this feedback may find weak or misleading relationships.

### 9.2 Equilibrium indeterminacy

When the agent's information is only moderately precise, multiple equilibria may exist. This makes the relationship between prices and actions unstable across settings and difficult to estimate with a single reduced-form specification.

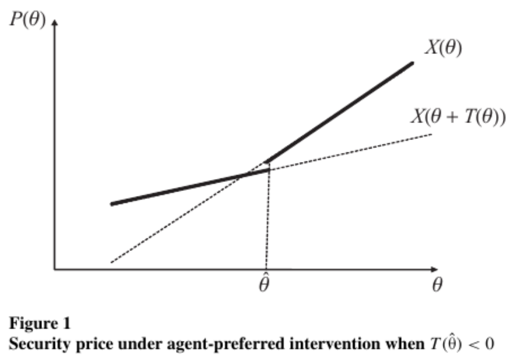
### 9.3 Security-specific predictions

The same fundamentals and intervention technology can produce different empirical patterns depending on the traded security. Equity prices may be associated with overintervention, while debt prices may be associated with underintervention. Therefore empirical tests should pay attention to whether the agent observes equity, debt, CDS, or multiple securities.

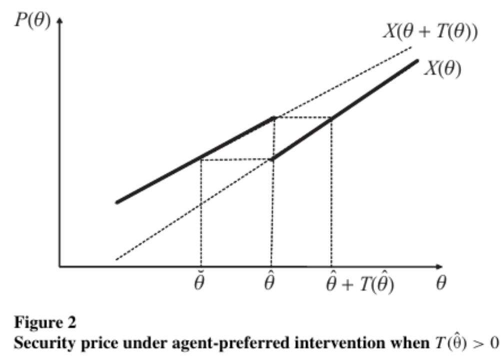
# 10 Tables and figures

## 10.1 Figure 1: Security price under agent-preferred intervention when $T(\hat{\theta}) < 0$

## 10.2 Figure 2: Security price under agent-preferred intervention when $T(\hat{\theta}) > 0$




(a) **Figure 1: Security price under agent-preferred intervention when $T(\hat{\theta}) < 0$.** Monotone price case.




(b) **Figure 2: Security price under agent-preferred intervention when $T(\hat{\theta}) > 0$.** Nonmonotone price case.

**Figure 1 – Read:**

- The vertical axis is the security price $P(\theta)$ and the horizontal axis is the fundamental $\theta$.
- The solid line $X(\theta)$ is the security value without intervention.
- The dashed line $X(\theta + T(\theta))$ is the security value with intervention.
- At the intervention cutoff $\hat{\theta}$, intervention lowers the security value: $T(\hat{\theta}) < 0$.
- The bold pricing function is monotone, so each price corresponds to one fundamental.

**Figure 1 – Interpretation:** When intervention reduces security value at the cutoff, the price remains an unambiguous signal of fundamentals. The agent can infer the state from the price and choose the preferred action. This is why Proposition 1 gives uniqueness and agent-preferred intervention for any signal precision.

**Figure 1 – Economic message:** Market-based actions are easy when the action does not create an upward jump in security value at the threshold. The price remains a clean signal.

**Figure 2 – Read:**

- In this case $T(\hat{\theta}) > 0$, so intervention raises security value at the cutoff.
- The bold price function jumps from the no-intervention branch to the intervention branch as fundamentals cross the cutoff.
- The same price can be generated by a low fundamental with intervention or a higher fundamental without intervention.
- The figure labels three connected points: $\check{\theta}$, $\hat{\theta}$, and $\hat{\theta} + T(\hat{\theta})$.

**Figure 2 – Interpretation:** This is the paper's central difficulty. When intervention is corrective, the price function becomes nonmonotone. Market prices do not reveal fundamentals by themselves. The agent needs his own signal to distinguish which side of the cutoff the firm is on.

**Figure 2 – Economic message:** Corrective intervention can make market prices less informative precisely because the market anticipates the correction. A higher price may reflect better fundamentals or expected intervention.

### 10.3 Figure 3: Security price in an equilibrium with too much intervention

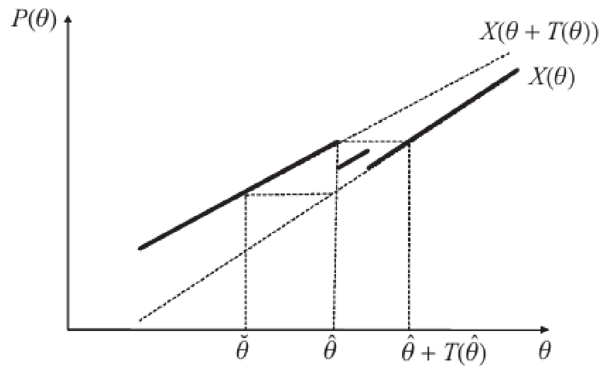



**Figure 3**
**Security price in an equilibrium with too much intervention**

**Figure 3: Security price in an equilibrium with too much intervention**. Overintervention equilibrium for a convex security.

**Read:**

- The figure shows an equilibrium different from agent-preferred intervention.
- The left branch corresponds to low fundamentals where intervention is desirable.
- The right branch corresponds to high fundamentals where no intervention is desirable.
- The middle branch lies above the cutoff but still involves positive intervention probability.
- Because the middle branch can share a price with a left-branch fundamental, the agent sometimes cannot distinguish the two states.

**Interpretation:** Figure 3 illustrates how overintervention can be self-confirming. If the agent is expected to intervene at some fundamentals above the cutoff, the price rises in those states. That makes those states harder to distinguish from low-fundamental states where intervention is desirable, so intervention above the cutoff becomes an equilibrium outcome.

**Economic message:** For a convex security such as equity, the price-feedback mechanism can lead agents to intervene too often. This shows that market-based governance or regulation can have real distortions even when markets are informative.

## 11 Short summary

- The paper studies agents who use market prices to choose corrective actions.
- Prices both inform and reflect expected action, creating a feedback loop.
- If intervention reduces value at the cutoff, prices remain monotone and fully revealing.

- If intervention is corrective and raises value at the cutoff, prices can become nonmonotone.
- With precise agent information, the agent can combine price and private signal to implement the preferred rule.
- With moderate information, multiple equilibria can arise; convex securities can generate too much intervention.
- With imprecise information, agent-preferred intervention fails and rational expectations equilibrium may not exist.
- Observing multiple securities, disclosing the agent's information, or using prediction markets can improve learning from prices.
- The central normative lesson is that market information and private monitoring are complements, not substitutes.

# 12 Conclusion

## 12.1 Core conclusion

Bond, Goldstein, and Prescott show that market prices are not always reliable triggers for corrective action. When an action is expected to improve firm value, the price incorporates that expectation, and this can make the price nonmonotone in fundamentals. The agent may then fail to infer the state correctly.

## 12.2 Economic intuition

The key intuition is that prices are forward looking. If a bad state triggers a valuable intervention, the price in that bad state may not be very low. This confounds inference. A market price can signal bad fundamentals plus expected correction or moderate fundamentals without correction.

## 12.3 Incremental contribution revisited

The paper's main contribution is to characterize rational expectations equilibria in this self-referential environment. It identifies when prices remain useful, when multiple equilibria appear, and when equilibrium can break down.

## 12.4 Policy implication

Market-based regulation or governance should not be used mechanically. Regulators, boards, activists, and managers need independent information to interpret prices. Policy designs that use multiple securities, disclosure, or prediction markets can help, but each has limitations.

## 12.5 Limitations

The model is theoretical and intentionally stylized. The market observes the fundamental perfectly in the baseline model; the agent's signal has bounded uniform noise; intervention is binary; and trading itself is represented through a rational-expectations price rather than a full microstructure model. These assumptions clarify the mechanism but limit direct quantitative application.

### 12.6 Takeaway

The enduring takeaway is that the usefulness of market prices depends on the action that prices are expected to trigger. If prices are used for corrective action, then the action changes what prices mean. Market discipline works best when decision makers bring their own information to the interpretation of the market signal.